

Bibek Gupta

Senior Software Engineer | Technical Lead | .NET, Azure & System Architecture

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SUMMARY

Senior Software Engineer and Technical Lead with 8+ years building production SaaS, ERP, and enterprise systems in C#, .NET Core, SQL Server, and Microsoft Azure. Currently leading a team of 7-8 engineers rewriting ERISApedia, an ERISA compliance-research platform now part of Ascensus, from legacy PHP to .NET Core, Angular, and MySQL on Azure and GCP, and shipping its AI Search (Ask ERISA) and Form 5500 data mining. Previously Senior Engineer and Team Lead on PensionPro, a SaaS platform used by 400+ third-party administrator firms managing 230,000+ U.S. retirement plans. Also built ForecastPro, an AI reporting SaaS in production with retail pharmacy chains in India.

TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript, SQL, PHP

Backend: .NET 8, .NET Core, ASP.NET Core Web API, Entity Framework Core, Dapper, Node.js, NestJS, Express, REST APIs, YARP API gateway, WebSockets, SignalR, background jobs (Quartz.NET, BullMQ)

Architecture: System design, Clean Architecture, microservices, event-driven architecture, serverless, multi-tenant SaaS, legacy modernization, API gateways, distributed locking (Redlock), API design, RBAC, high availability

Cloud & DevOps: Microsoft Azure (Functions, Logic Apps, Service Bus, Queue Storage, Table Storage, Event Grid, App Service, Azure AD), Google Cloud Platform (Cloud Run functions, Cloud Tasks, Cloud Storage, BigQuery), Docker, Azure DevOps, GitHub Actions, CI/CD, Linux, Nginx

Databases: SQL Server, PostgreSQL, MySQL, Redis, Elasticsearch, BigQuery, read replicas, query optimization

Frontend: Angular, React, Next.js, TypeScript, Tailwind CSS, PrimeNG, Ag-Grid

Security & Integrations: SAML 2.0 SSO, OAuth 2.0 / OpenID Connect, JWT, AES encryption, Stripe, DocuSign, QuickBooks Online, Braintree, Zendesk, Google Sheets & Drive APIs

AI & Data: LLM integration (OpenAI-compatible APIs), LLM-powered search, natural language to SQL (NLQ), semantic layer design, token metering, time-series forecasting

Testing: Unit and integration testing, Jest, Vitest, Playwright

Leadership: Team leadership, code review, design review, coding standards, mentoring, requirements analysis, collaboration with U.S. stakeholders

PROFESSIONAL EXPERIENCE

Technical Lead / Senior Software Engineer | Jun 2022 - Present

DolphinDive Technology (client: AmericanTCS, acquired by Ascensus in 2026) | Kathmandu, Nepal

Engineering team for U.S. retirement-industry SaaS products (PensionPro, ERISApedia), now part of Ascensus, which serves 16M+ savers and \$930B+ in assets under administration.

ERISApedia - Team Lead | Nov 2025 - Present

- Lead a team of 7-8 engineers on the ground-up rewrite of ERISApedia, an ERISA compliance-research platform, replacing a legacy PHP application with .NET Core, EF Core, Angular, MySQL, and Redis.
- Team delivered AI Search (Ask ERISA), launched July 2026: LLM-powered answers across nine ERISA compliance book titles for advisors, auditors, and TPAs.
- Team built Form 5500 and DOL data pipelines that scrape and mine public filings on Google Cloud (Cloud Run functions, Cloud Tasks, Cloud Storage, BigQuery), searchable on hundreds of criteria with benchmarking.
- Own the target architecture, technology selection, and migration path off the legacy system, and the cloud design across Azure and GCP, working with U.S. stakeholders on scope and sequencing.
- Made the new platform faster than the legacy PHP system by rewriting UI components, optimizing slow SQL queries, and adding Redis distributed caching at the API layer.

PensionPro - Senior Software Engineer, then Team Lead (2 years) | Jun 2022 - Nov 2025

- Major contributor to the .NET Core architecture of a pension administration SaaS used by 400+ TPA firms administering 230,000+ retirement plans, and its companion apps Fetch and PlanSponsorLinks.
- Designed and architected a YARP-based API gateway for third-party consumers of the platform's external APIs, handling their authentication and rate limiting with Redis distributed locks (Redlock), mutexes, and semaphores.
- Worked on a read path built for scale: Redis distributed caching, read-only database replicas, a search-only database, and Elasticsearch.

- Moved long-running and integration-heavy work off the request path using Azure Service Bus, Storage queues and Table Storage, Azure Functions (all trigger types), Logic Apps, and Event Grid, so background processing scales independently of user traffic.
- Integrated ERISApedia, ftwilliam.com, DocuSign, and Form 5500 data, plus Zendesk, QuickBooks Online, and Braintree; implemented SAML 2.0 single sign-on and identity management.
- Built and maintained Azure DevOps build (CI) pipelines, working with the DevOps team that owns release pipelines on a 2 to 4 week production release cadence.
- As Team Lead, set code review, design review, and coding standards and mentored up to 4 engineers plus QA at a time; served as primary engineering liaison translating pension-compliance requirements into technical designs.

Senior Software Engineer (Freelance Consultant) | Jun 2022 - Jun 2025 **Bharuwa Solutions (India)**

Enterprise software products: ERP, fintech, distribution, and agritech platforms.

- Architected QueryGen, a domain-driven ASP.NET Core engine used in production that generates dynamic queries and reports with per-client database routing, eliminating hand-written SQL for routine reporting.
- Re-engineered ERP accounting, payroll, and invoicing modules for multi-tenant operation, letting one deployment serve multiple client organizations with isolated data.
- Automated GST reporting and filing against the government e-portal, replacing a manual compliance process.
- Built inventory and logistics tracking for a Warehouse Management System with multi-level category hierarchies.

Software Engineer | Jun 2019 - May 2022

IMS Software Pvt. Ltd. | Kathmandu, Nepal

Retail and distribution platforms used by 20,000+ businesses on 100,000+ POS terminals across Nepal, India, and Southeast Asia.

- Engineered a custom ERP for Patanjali Ayurved (India), the company's flagship cross-border enterprise implementation on one of India's largest FMCG distribution networks.
- Built and owned production retail POS and distribution management systems: inventory, customers, sales reporting, and multi-level distribution hierarchies.
- Integrated accounting workflows into the POS and administered the production Linux servers behind it.

Software Engineer | Apr 2018 - Mar 2019

Softweb Developers | Kathmandu, Nepal

- Designed the front-end architecture and a reusable component library shared across applications.
- Optimized relational schemas for query performance and added WebSocket-based real-time features.

PROJECTS

ForecastPro - Multi-tenant forecasting & AI reporting SaaS | Sole engineer, in production

- In production with a few large retail pharmacy chains in India: syncs their ERP data through a resumable pipeline, forecasts demand, and answers business questions asked in plain English.
- Built a natural-language reporting (NLQ) engine: an LLM maps questions to a catalog-constrained semantic query that is compiled to parameterized SQL with tenant and branch row-level security and SQL safety checks, so the model never writes SQL.
- Built an AI insights engine with 12 pluggable providers (stock-out risk, churn, dead stock, discount anomalies, business health score), Redis-cached dashboards, and scheduled generation.
- Built prepaid AI credit billing: append-only ledger enforced by database triggers, reservation-based charging, per-tenant pricing and spend limits, and signature-verified, idempotent Stripe webhooks.

SastoRent - Rental and services marketplace | Freelance, lead engineer

- Built a marketplace (sastorent.com) where people rent out properties and everyday items, and where service providers such as plumbers, carpenters, and tutors register and sell their services.

LearnSphere - Multi-tenant learning management system | Lead engineer

- SAML and OpenID Connect SSO, Stripe payments, queue-based exports, Redis caching, attendance and marksheets, and Prometheus metrics across 112 database tables; deployed with Docker.

DriveNow and Futsal Booking - Booking platforms | Freelance

- Car rental and futsal court booking platforms with online payments, availability checks that prevent double bookings, role-based admin panels, and customer dashboards.

EDUCATION

BSc Computer Science | Tribhuvan University | 2018